Vehicle Speed and Density Estimation Using Deep Learning on CCTV Feeds

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF CSE

Name of Title: Vehicle Speed and Density Estimation Using Deep Learning on CCTV Feeds

1. Abstract:

Vehicle speed and traffic density estimation are important components of modern intelligent transportation systems, helping to improve road safety, traffic management, and urban mobility. Traditional methods of monitoring vehicle speed and density often depend on manual observation, roadside sensors, or specialized equipment, which can be expensive, time-consuming, and difficult to deploy on a large scale. This project, **"Vehicle Speed and Density Estimation Using Deep Learning on CCTV Feeds,"** proposes an automated computer vision-based system that analyzes CCTV camera footage to estimate vehicle speed and traffic density in real time. The proposed system uses deep learning and image processing techniques to detect and track vehicles from CCTV video streams. A suitable object detection model, such as YOLO, can identify different vehicles including cars, buses, trucks, and motorcycles. After detection, vehicle tracking techniques are applied to follow individual vehicles across consecutive video frames. By analyzing the movement of vehicles between predefined regions in the video and using calibrated camera parameters, the system estimates the approximate speed of each detected vehicle. At the same time, the number of vehicles present within a particular road region is calculated to determine traffic density. The system can classify traffic conditions into categories such as low, medium, and high density, providing useful information for traffic monitoring and management. The estimated speed and density values can be displayed through a user-friendly dashboard, allowing authorities to understand traffic conditions quickly. The proposed approach reduces the need for manual monitoring and can operate continuously using existing CCTV infrastructure.

2. INTRODUCTION:

With the rapid increase in the number of vehicles in urban areas, traffic congestion, speeding, and road safety have become major challenges. Traditional traffic monitoring methods require manual observation or additional sensors, which can be costly and time-consuming. Vehicle Speed and Density Estimation Using Deep Learning on CCTV Feeds provides an automated solution by using existing CCTV cameras to monitor road traffic. The system uses deep learning and computer vision techniques to detect and track vehicles from video footage.

The proposed system estimates the speed of vehicles by analyzing their movement across consecutive video frames and determines traffic density by counting the number of vehicles within a specific road area. Deep learning models such as YOLO can be used for accurate and fast vehicle detection. The system can classify traffic conditions as low, medium, or high density and provide useful information for traffic management. This approach can reduce manual monitoring and support applications such as smart cities, traffic control, congestion analysis, and road safety monitoring. Overall, this project demonstrates how deep learning and computer vision can be integrated with CCTV feeds to create an efficient, scalable, and cost-effective traffic monitoring solution. The system has potential applications in smart

cities, traffic control centers, road safety monitoring, congestion analysis, and intelligent transportation systems.

3. SOFTWARE REQUIREMENTS:

The proposed **Vehicle Speed and Density Estimation Using Deep Learning on CCTV Feeds** system requires both hardware and software components to perform vehicle detection, tracking, speed estimation, and traffic density analysis efficiently. A computer with sufficient processing power is needed to execute deep learning models and process CCTV video streams in real time. The software environment includes Python as the programming language, deep learning frameworks, computer vision libraries, and development tools for implementing and testing the system. Existing CCTV cameras serve as the video input source, making the solution cost-effective by utilizing already installed surveillance infrastructure.

Category	Requirement	Specification
Hardware	Processor	Intel Core i5 / AMD Ryzen 5 or above
Hardware	RAM	8 GB (16 GB recommended)
Hardware	Storage	256 GB SSD or higher
Hardware	Graphics	NVIDIA GPU (optional, recommended for faster deep learning processing)
Hardware	Camera	CCTV Camera/IP Camera
Hardware	Display	Monitor with 1080p resolution
Software	Operating System	Windows 10/11 or Ubuntu Linux
Software	Programming Language	Python 3.10 or above
Software	IDE	Visual Studio Code / PyCharm / Jupyter Notebook
Software	Deep Learning Framework	TensorFlow or PyTorch
Software	Computer Vision Library	OpenCV
Software	Object Detection Model	YOLO (You Only Look Once)
Software	Database (Optional)	MySQL or SQLite

Signature of the Guide **Course Instructor**